Egocentric Laundry Folding: Replaying Hand-Worn-Gripper Demonstrations on a Bimanual Robot

Jeong Hwan Lee
Technical report, version 0, 21 October 2026
Repository: `egocentric-laundry-folding`

***Abstract*—We want a two-arm robot to fold a T-shirt from human demonstrations recorded with hand-worn grippers, without teleoperation, and first ask whether replaying the converted human motion makes the robot do what the human did. The demonstrator wears two HandUMI grippers (an open-source design by BrikHMP18, not ours), each carrying three AprilTags, a wrist fisheye camera and an IMU; a fixed RGB-D camera observes the tags. A camera-to-robot calibration (0.98 px held-out), start markers measured in the robot base frame and a calibrated tag rig (1–4 mm static error) give each jaw-tip TCP in the robot base frame without per-take alignment, and a classifier on the wrist video gives the jaw state. The human path is robotized by smooth warps, planned with joint-limit-aware IK and safety checks, and replayed on two reBot B601-DM arms through an MIT-mode bridge, with every sensor and the robot log on one host clock. Single-arm replays pass 16/18 translation strokes (median per-axis RMSE 5.5–11.6 mm) and the rotation strokes, and the right arm grasped and lifted the T-shirt from human data. Full two-arm replays of a 33 s demonstration (94 s at 0.35×) run reliably: the jaws close 5–19 mm above the table (human ≈10 mm), the robot stays within 2–8 mm of its command at each grasp, and runs repeat within about 1 mm. The robot does not yet fold like the human: the position error against the replayed (robotized) human path is 17 mm (right) and 30 mm (left) RMSE, open-loop replay needs the cloth where the human's was, and wrist-camera SLAM was rejected as the trajectory source because it loses tracking exactly at grasps. We report no success rate.**

I. Introduction

Hand-held and hand-worn grippers in the style of UMI [1] let a person demonstrate with an end-effector that looks like the robot's, without a robot in the loop. Our goal is a two-arm robot that folds a T-shirt on a table from such demonstrations, with human data as the main source rather than teleoperation.

Before training a policy on such data, we check a narrower thing: *if the robot simply replays the converted human motion, does it do what the human did?* If replay fails, a policy trained on the same labels inherits the failure. Replay therefore serves as an end-to-end test of calibration, pose recovery, jaw labelling and conversion (Fig. 1). The demonstrations are recorded with two HandUMI grippers [2], [3] and replayed on two low-cost 6-DoF arms. This report describes what was built and measured up to 21 October 2026. Its contributions are:

- one world frame, the robot base frame, set once by a camera-to-robot calibration, start markers and a per-gripper tag rig, so that human and robot paths share it without per-take alignment (Sec. V);
- a conversion from the recording to a robot-feasible jaw-tip TCP path with jaw labels: tags plus depth, IMU gap filling, robotization by continuous warps with a per-frame quality label, and a bimanual planner with explicit safety checks (Sec. VI);
- a replay and evaluation setup in which human, command and robot are on one clock and filmed by the same camera, and the measured results, including the methods that failed (Secs. VII–VIII).

Roadmap. Section II reviews related work and Sec. III gives an overview of the data flow from the UMI recording (D0) to robot motion (D7). Section IV describes the hardware, Sec. V the one-time calibration that defines the shared world frame, and Sec. VI each transformation of the chain. Sections VII and VIII give the experimental setup and results, and Secs. IX–XI the lessons, limitations and next steps. Appendix A lists the commands in data-flow order and Appendix B the repository layout.

II. Related Work

A. Demonstration interfaces and human data

Hand-held and hand-worn grippers. UMI [1] records demonstrations with a hand-held parallel-jaw gripper carrying a wrist fisheye camera; the end-effector pose comes from a monocular-inertial SLAM system based on ORB-SLAM3 [18], the finger width from fiducial markers, and the trajectories are filtered by the kinematics of the target robot. UMI also matches sensor and actuation latencies at inference and uses a relative-trajectory action representation. HandUMI [2], [3], designed and published by BrikHMP18 and co-authors, is an open-source hand-worn variant of UMI for robot-free bimanual data capture with swappable parallel-jaw tips; we use a build copy of it (Sec. IV). DexUMI [4] uses the human hand with a wearable exoskeleton as the interface for dexterous robot hands and closes the visual gap by inpainting the robot hand into the video. Our work keeps the UMI-style gripper but replaces wrist-camera SLAM by absolute fiducial tracking from a fixed RGB-D camera, and it tests the data by open-loop replay on the target robot before any policy training. UMI's kinematic filtering removes infeasible trajectories; we

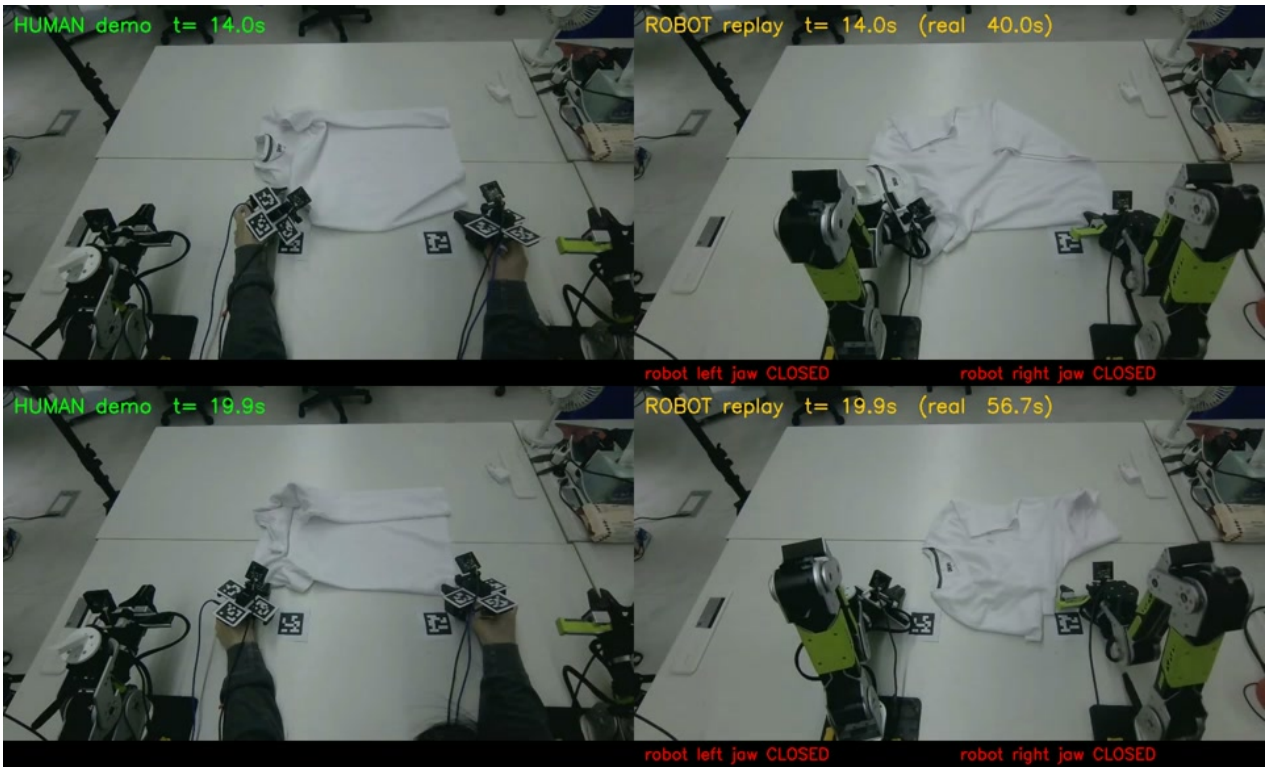


Fig. 1. Human demonstration (left) and robot replay of the converted trajectory (right), filmed by the same fixed Orbbec camera and aligned on one clock (take `20261021_171057`, demo 2; run `bi_20261021_200335`; robot at 0.35×). Rows: robot time 40 s and 57 s (human time 14 s and 20 s); the bars show the robot jaw commands. The jaws hold the shirt edge and drag it. Full videos: `media/`.

instead warp the human path into a feasible one and store how far each frame was moved (Sec. VI-F).

Human and egocentric video. EgoMimic [5] trains a single policy on egocentric human video with 3D hand tracking together with robot demonstrations. Our long-term aim is similar (human data as the main source), but this report covers only the step that turns hand-worn-gripper recordings into robot-frame end-effector and jaw labels and checks them by replay.

B. Robots and cloth

Low-cost bimanual robots. ALOHA [6] showed fine-grained bimanual manipulation with low-cost arms and teleoperated demonstrations. We use two low-cost reBot B601-DM arms but record without teleoperation. The robot bridge runs on a reBot plugin for LeRobot [7].

Cloth manipulation. Garment folding and unfolding have been learned with dedicated primitives, e.g. bimanual folding in SpeedFolding [8] and dynamic flinging in FlingBot [9]. We do not propose a cloth method; the cloth enters our results only as the object the replayed grasps must reach.

C. Pose estimation, calibration and kinematics

Fiducial markers. We use AprilTag [10], [11] tag36h11 markers detected with OpenCV [12]. A single square tag is solved with IPPE [13], which returns the two candidate poses of a planar target; when several tags of one rig are visible, all corners are solved jointly with SQPnP [14] followed by Levenberg–Marquardt refinement.

Visual and visual-inertial SLAM. To fill the tag-hidden grasp phases we tested MASt3R-SLAM [15], a dense monocular SLAM system built on the MASt3R [16] and DUSt3R [17] two-view reconstruction priors; ORB-SLAM3 [18] in monocular-inertial mode; and, with an earlier stereo camera, Basalt [19], OpenVINS [20] and AirSLAM [21]. Section VIII-F1 reports why none of them became the trajectory source.

Calibration and kinematics. Camera–IMU calibration of the earlier pipeline used Kalibr [22]. The camera-to-robot transform is initialised with the robot-world/hand-eye solution of Shah [23] (as implemented in OpenCV) and refined by our own least squares. Inverse kinematics uses Pink [25] on top of Pinocchio [24]. The jaw classifier is a ResNet-18 [26] pretrained on ImageNet [27].

III. System Overview

Figure 2 shows how a demonstration becomes robot motion as a chain of data artifacts D0–D7. Two principles run through it: every pose lives in one world frame, the robot base frame, and every sensor and robot log is stamped by one host clock.

- *Setup with AprilTag markers* (yellow, once; Sec. V): a calibration marker on the robot wrist gives the camera pose in the robot base, two start markers on the table are measured in robot coordinates, and the three markers on

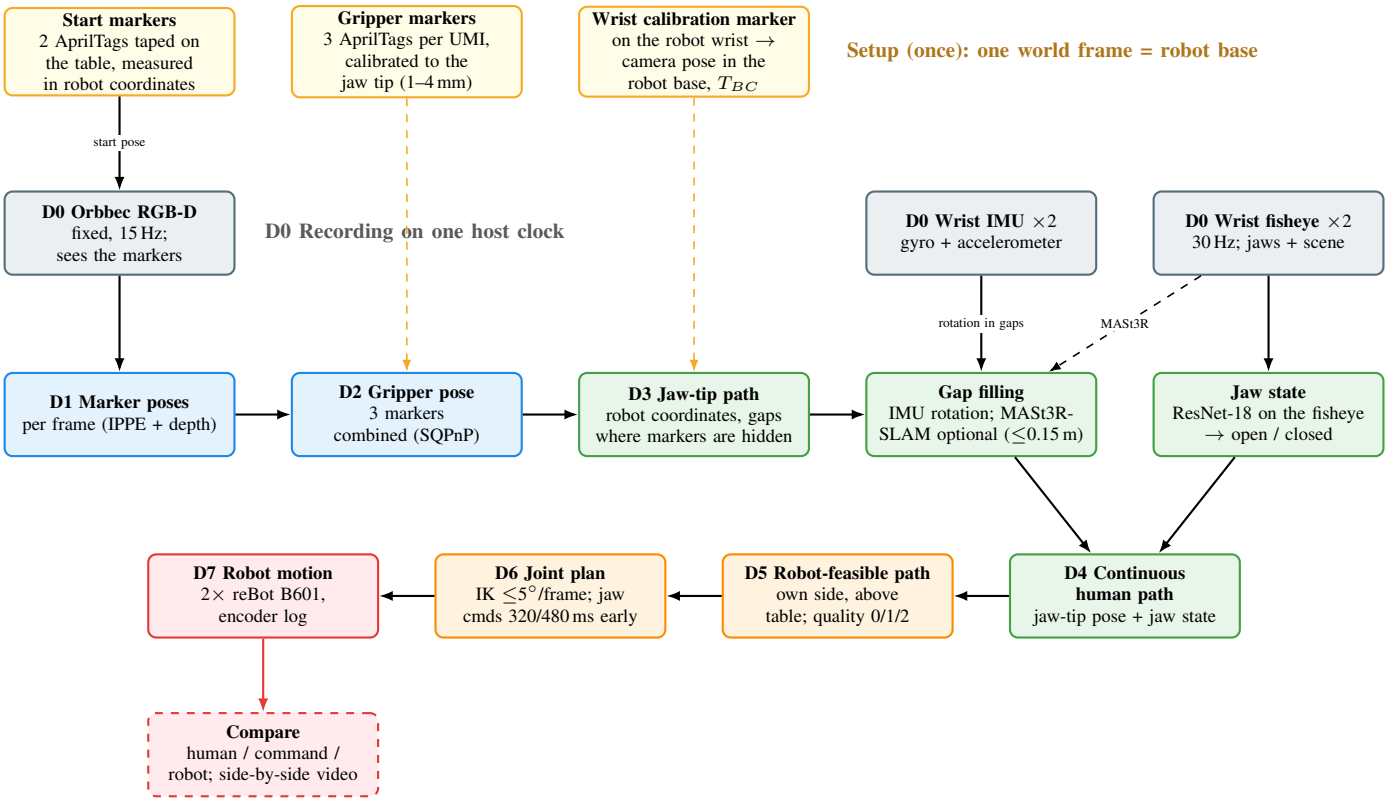


Fig. 2. Data flow from the UMI recording to robot motion. Yellow: one-time setup with AprilTag markers that puts human and robot in one world frame (the robot base). Grey: the three sensors of the recording D0. Blue: marker and gripper poses in the camera frame. Green: the human jaw-tip path in robot coordinates, completed by IMU gap filling (MASt3R-SLAM optional) and the jaw state from the wrist fisheye. Orange: robot-feasible path and joint plan. Red: robot and comparison. Dashed: calibration inputs and the optional SLAM filler.

each gripper are calibrated to its jaw tip. Every take starts with the grippers at the start markers.

- *Recording D0* (grey; Sec. VI-A): three sensors on one clock. The fixed Orbbec RGB-D camera sees the markers and is the main position source; the wrist IMU supplies rotation while the markers are hidden; the wrist fisheye shows the jaws (jaw state) and the scene (optional MASt3R-SLAM).
- *Marker and gripper poses* (blue): per-frame marker poses D1 (Sec. VI-B) and the gripper pose D2 (Sec. VI-C), in the camera frame.
- *Human path in robot coordinates* (green): the jaw-tip path D3 (Sec. VI-D); IMU gap filling, the optional MASt3R filler and the jaw state make it the continuous path D4 (Sec. VI-E).
- *Robot-feasible path and plan* (orange): robotization gives D5 (Sec. VI-F); IK gives the joint plan with jaw commands D6 (Sec. VI-G), executed only if the safety checks pass (Sec. VI-H).
- *Robot* (red): the bridge executes the plan and logs the robot motion D7 (Sec. VI-I), which is compared with D4/D5 and filmed by the same camera (Sec. VII).

IV. Hardware

Table I lists the components.

TABLE I
Components

unit	model, role
Hand-worn gripper ×2	HandUMI [2] (BrikHMP18), our build; human demonstration tool
Wrist camera ×2	fisheye, 1280×720 at 30 Hz (left measured 24 fps in take 171057); jaw state, optional SLAM
Wrist IMU ×2	Seeed XIAO nRF52840 Sense; gyro and accelerometer
RGB-D camera	Orbbec Gemini 336, colour + aligned depth at 15 Hz, fixed ≈0.8 m above the table
Robot arms ×2	reBot B601-DM, 6-DoF + parallel jaw, Damiao motors on USB-CAN
Host	Apple-silicon laptop: recording, planning, bridge
GPU (optional)	RTX 4090/5090 nodes: MASt3R-SLAM

A. HandUMI and our modifications

HandUMI [2], [3] is a hand-worn parallel-jaw gripper designed and published by BrikHMP18 (Apache-2.0). It is not our design. We print the upstream parts (commit `e58de33`) and changed the following (Fig. 3):

- three modified parts: a controller-support part turned into a connector with a +10 mm plate (we use no VR

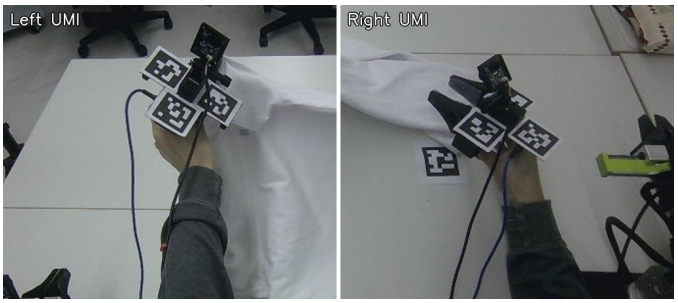


Fig. 3. The UMI hands as used now (cut from a demo recording): HandUMI with three AprilTags per hand (left UMI on the left, IDs 210–212; right UMI on the right, IDs 200–202), wrist fisheye and IMU; no jaw servo.

controller), a camera mount with an extra window, and a servo cover with a wider cable opening;
- a wrist IMU instead of VR tracking (first a Teensy 4.1 with an ICM-42688-P, now a Seeed XIAO nRF52840 Sense per hand);
- the jaw servos removed on 10-13, because the servo shafts broke and the units were heavy;
- the red jaw tape removed on 10-20, because the colour detector fired on background objects;
- three tag36h11 AprilTags per hand (48 mm black square, on different faces so that at least one faces the camera in most poses).

Both wrist fisheye cameras report the same serial number, so they are identified by USB location.

B. Scene and robot arms

The Orbbec Gemini 336 is fixed about 0.8 m above the table (camera origin at $z = 0.79$ m in T_{BC}; its view is Fig. 4). Two reBot B601-DM arms are mounted 560 mm apart (tape; the URDF assumes 630 mm). Two *start markers* (AprilTags with IDs 400 and 401, 60 mm) are taped on the table where the left and right grippers are placed at the start of every take, and a *wrist calibration marker* (AprilTag ID 220, 50 mm) is fixed on the wrist-roll motor body of the robot so the camera can locate the robot. The gripper zeros drift at every power cycle and are read before each session; the left jaw travel is limited to half.

C. Robot bridge

Our HTTP service drives both arms from one 100 Hz thread with synchronised, acceleration-limited motion toward each goal (45°/s, 180°/s^2), goal interpolation between streamed commands, and parking with torque off at rest.

Control law. In MIT mode each drive receives a position and velocity reference, gains, and a feed-forward torque. The bridge uses an integral-only feed-forward law and no gravity model:

$$\tau = K_p\, e + K_d\, \dot{e} + \tau_{\text{ff}}, \quad e = q^\star - q, \tag{1}$$

$$\tau_{\text{ff}}[k+1] = \tau_{\text{ff}}[k] + K_i\, e[k]\, \Delta t. \tag{2}$$

While the arm moves, the joint torque must follow the changing gravity load τ_g. In quasi-steady motion ($\dot{e} \approx 0$), differentiating the torque balance $\tau_g = K_p e + \tau_{\text{ff}}$ gives $\dot{\tau}_g = K_p \dot{e} + K_i e$, hence

$$e \approx \dot{\tau}_g / K_i, \tag{3}$$

independent of K_p. This matched the measurements: raising K_p on j2/j3 from 64 to 96 did not change the moving error, and a simulation of gravity plus the integral law explained 77% of the j2 error. The gains were therefore tuned through K_i (Sec. VIII-B); j3 stayed at 56 because with these gains it already overshot by 0.47°. The deployed gains are $K_p = (64, 96, 96, 28, 28, 28)$ and $K_i = (56, 84, 56, 56, 56, 56)$ N·m/(rad·s). The known cost of the law is a sag when a loaded joint first moves, until the integral winds up.

Fault recovery. Two recovery paths were added after hardware faults on 10-21: a motor that reports "disabled" is re-enabled in place and the arm holds its measured pose (the left wrist connector has a bad contact); after a USB "broken pipe" the arm's session is rebuilt, and the bridge waits up to 30 s for the USB location to reappear. During a replay, streaming pauses while one arm's reads are stale and resumes when they return; an operator E-stop stops at once.

V. One World Frame for Human and Robot

A. Overview: how the world frame is set up

Human and robot data use one world frame: the robot base frame B of the right arm. It is set up in four steps, all once per installation:

1) *Camera to robot.* The robot moves the wrist calibration marker through 32 poses in front of the fixed Orbbec, which gives T_{CB} (Sec. V-C). From then on every pose the camera measures can be written in B: $T_{B\cdot} = T_{BC}\, T_{C\cdot}$ with $T_{BC} = T_{CB}^{-1}$. The left arm base is located in the same frame by a second solve (Sec. V-D).
2) *Start markers.* The two start markers on the table are measured once with the calibrated camera (PnP + depth, 51 frames; Fig. 4a, Table II).
3) *Gripper markers.* The three markers of each gripper are calibrated to its jaw tip (Sec. V-E), so the camera gives the jaw-tip position directly in B.
4) *Start of every take.* Each take starts with the grippers at the start markers (Fig. 4b), so every demonstration begins at the same known robot coordinates.

No per-take alignment (no SLAM map, no point-cloud registration) is needed: a camera measurement of a gripper marker is already a robot-frame pose.

The one-time calibration is the side input of the data chain (Fig. 2): it defines the shared world frame, the robot base frame, and enters the chain at D1→D2, D2→D3 and the IK. Table II collects the results; their accuracy is discussed in Sec. VIII-A.

B. Notation

$T_{XY} \in SE(3)$ maps coordinates in frame Y to frame X; R_{XY} and t_{XY} are its rotation and translation. Frames: B robot

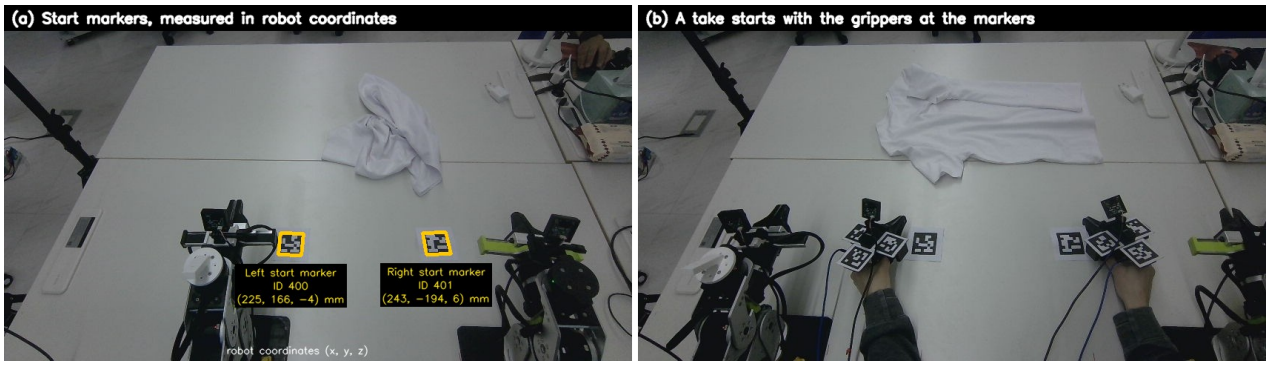


Fig. 4. Start markers. (a) Two AprilTags taped on the table (IDs 400 left, 401 right, 60 mm), measured once with the calibrated Orbbec; their centres in robot coordinates are shown. (b) Every take starts with the two grippers at these markers, so each demonstration begins at known robot coordinates.

TABLE II
CALIBRATION RESULTS

item	result
Orbbec ↔ robot base	18/32 poses, held-out 0.98 px, 1.3 mm lateral; human vs robot axes 1–6°
Robot TCP	URDF tip 89 mm off → −0.019 m from gripper link; READY tip within 6–9 mm of tape
Left arm base	70 mm inward, ≈3° tilt (1.6 px, held-out 1.9 px)
Left j2	−8.4° offset, 3 poses within 5–10 mm
Start markers 400/401	centres $(225, 166, -4)$ mm and $(243, -194, 6)$ mm
UMI tag rig → jaw tip	static 1–4 mm, leave-one-out 1.8–4.8 mm
UMI → robot gripper orientation	spread R 2.3°, L 3.8–8.6°
IMU ↔ Orbbec (rig)	time offset R 15 / L 10 ms
Wrist cam ↔ IMU	offset R −40 / L −35 ms; rotation from data
Wrist cam → TCP	translation by tape, rotation solved

base, C Orbbec colour camera, G gyro integration frame, I IMU, E robot TCP, and per UMI a rig frame (tag 200 or 210) with tags i. The robot base frame follows the reBot URDF of the right arm: $+X$ forward, $+Y$ left, $+Z$ up, table at $z_{\mathrm{t}} = 3$ mm. The jaw-tip TCP is the midpoint between the jaw tips on both sides, for the UMI and for the robot. The jaw signal is $g = 1$ open, $g = 0$ closed.

C. Camera to robot base

With the wrist calibration marker (ID 220) on the robot, the robot holds 32 poses q_k around READY while the Orbbec records. For each hold, the tag corners are taken as the median over the frames of the hold. The unknowns T_{CB} and $T_{E,220}$ satisfy

$$T_{C,220}(k) = T_{CB}\, T_{BE}(q_k)\, T_{E,220}, \tag{4}$$

where $T_{BE}(q_k)$ is the forward kinematics. They are initialised with the method of Shah [23] and refined by minimising the reprojection error of all tag corners (Huber, 2 px) plus a residual between the predicted tag depth and the aligned Orbbec depth; at about 1 m range the PnP depth of a 48–50 mm tag alone was too noisy to start from. The solution is checked by five-fold held-out validation. The right-arm T_{CB} defines the world frame.

D. Robot model corrections measured in the world

Three errors of the robot model were found only by measuring the real robot:

- *TCP:* the URDF gripper TCP (gripper link + 70 mm) lies 89 mm in front of the real jaw tips, found with a tape measure at READY (tip 360 mm forward of the base centre, 280 mm above the table). The TCP is set to −19 mm from the gripper link. Every earlier absolute replay put the real jaws short by this offset along the tool axis; FK-versus-FK parity could not see it.
- *Left arm base:* solving T_{CB} a second time with the tag on the *left* wrist showed the left base 70 mm inward of the URDF with a 3° tilt. It is applied as a correction D of the left `joint1` placement.
- *Left j2:* model joint = bridge joint + offset; a j2-only fit on READY and two further tape-measured poses gives −8.4°.

All three are applied as overrides inside the IK model; the URDF file is untouched.

E. UMI tag rig

Slow rotation takes give the tag-to-tag transforms $T_{\mathrm{rig},i}$ of each rig: each detection is solved with IPPE, and $T_{\mathrm{rig},i}$ is a robust mean over good co-visible frames (chordal mean of the rotations, median of the translations), chained through a common tag when two tags are never seen together. The jaw-tip position t_{tcp} in the rig frame comes from touching the corners of a board of known geometry (16 holds per hand):

$$t_{\mathrm{tcp}} = \arg\min_{t} \sum_{j} \rho\big(\| p_{B,\mathrm{rig}}(j) + R_{B,\mathrm{rig}}(j)\, t - c_B(j) \|\big), \tag{5}$$

with ρ a Huber loss (scale 5 mm) and $c_B(j)$ the touched corner. A pivot take (jaw tip held on one point while the hand is tilted by about 50°) gives a second, independent estimate.

F. Other calibrations

The UMI-to-robot-gripper orientation R_{UE}, used by the planner, comes from a take in which the UMI is held parallel to the robot gripper at READY. The remaining two are used only by the optional SLAM filler (Sec. VI-E). The time offsets between wrist camera and IMU and the camera–IMU rotation are estimated from data (MASt3R rotation rates against the gyro); the Kalibr rotation from September was 173° off for the current right mount. The wrist-camera-to-TCP translation is taped (right $(-5, 34, 122)$ mm, left $(-15, 14, 118)$ mm) and only the rotation is solved; a joint solve that gave a 159 mm optical-axis offset was rejected.

VI. From UMI Data to Robot Motion

This section follows the data chain of Fig. 2 for one take, one subsection per transformation.

A. Recording: demonstration → D0

Input: a human demonstration with both UMIs. *Output:* D0, the raw recording on one clock.

One recorder writes the Orbbec colour and aligned depth (15 Hz), both wrist videos (30 Hz) and both IMUs. A spoken protocol structures each take: the start pose with both UMIs at the start markers (Fig. 4b), then the demos (each: hold, 30 s of free motion, hold), then a final hold. Starting at the start markers puts every take in the shared world frame (Sec. V-A); the protocol cues define the demo windows used by all later transformations. Every stream is stamped with the host monotonic clock; IMU device time is mapped to host time by a fitted clock model. The replay later logs the commanded and the measured joints on the same clock, so no synchronisation hardware is needed.

B. Tag detection: D0 → D1

Input: the Orbbec colour and depth frames of D0. *Output:* D1, the pose of every visible tag per frame in the camera frame.

Tags with a pixel side under 18 px are ignored. Each remaining tag i is solved with IPPE, giving $T_{C,i}$ and its centre c_i. If the median Orbbec depth d_i inside the tag quadrilateral is valid and within 0.10 m of $c_{i,z}$, the centre is moved along its ray: $c_i \leftarrow c_i\, d_i / c_{i,z}$. At about 1 m range, depth is the least accurate PnP coordinate.

C. UMI tag rig: D1 → D2

Input: D1 and the calibrated tag rig $T_{\mathrm{rig},i}$ (Sec. V-E). *Output:* D2, the rig pose $T_{C,\mathrm{rig}}$ of each UMI per frame in the camera frame.

1) *Rig pose from the tags.* With one visible tag, $T_{C,\mathrm{rig}} = T_{C,i} T_{\mathrm{rig},i}^{-1}$. With two or more, all corners $X_{ik} = T_{\mathrm{rig},i}\, x_k$ are solved jointly,

$$T_{C,\mathrm{rig}} = \arg\min_{T} \sum_{i,k} \|\pi(K, T X_{ik}) - u_{ik}\|^2, \quad (6)$$

initialised with SQPnP and refined with Levenberg–Marquardt. Frames with reprojection RMSE above 2 px are dropped.

2) *Rotation.* Either from the tags, or from the gyro anchored to the tags:

$$R_{C,\mathrm{rig}}(t) = \bar{R}_{CG}(t)\, R_{GI}(t)\, R_{I,\mathrm{rig}}, \quad (7)$$

where R_{GI} integrates the gyro minus its bias (median over still samples), $R_{I,\mathrm{rig}}$ is calibrated, and $\bar{R}_{CG}(t)$ is the weighted rotation mean of $R_{C,\mathrm{rig}}^{\mathrm{tag}} R_{I,\mathrm{rig}}^{\top} R_{GI}^{\top}$ over tag frames within ±10 s (weight 1 with two or more tags, 0.15 with one).

3) *Position.* $p_{C,\mathrm{rig}} = \frac{1}{|V|} \sum_{i \in V} \left(c_i - R_{C,\mathrm{rig}}\, t_{\mathrm{rig},i}\right)$ over the visible tags V; using the fused rotation removes the leak of single-tag rotation error into position.

D. Jaw tip in the robot base frame: D2 → D3

Input: D2, the jaw-tip offset t_{tcp} (Sec. V-E) and T_{BC} (Sec. V-C). *Output:* D3, the human jaw-tip TCP trajectory in the robot base frame, with gaps where no tag is seen.

$$p_B = R_{BC}\left(R_{C,\mathrm{rig}}\, t_{\mathrm{tcp}} + p_{C,\mathrm{rig}}\right) + t_{BC}, \quad (8)$$

$$R_{B,\mathrm{tcp}} = R_{BC} R_{C,\mathrm{rig}}. \quad (9)$$

No per-take alignment is applied: the start markers place the start of every take in the robot base frame.

E. Gap filling and jaw state: D3 (+ D0) → D4

Figure 5 shows which source supplies each frame of a full demonstration. *Input:* D3, the IMU and the wrist video of D0. *Output:* D4, a continuous human trajectory with jaw-tip pose and jaw state per frame in the robot base frame.

IMU. Frames without a tag get the gyro-anchored rotation of Sec. VI-C and a linearly interpolated position, and are flagged ($n_{\mathrm{tags}} = 0$) as not measured.

Optional MASt3R-SLAM filler. MASt3R-SLAM [15] runs on the wrist video undistorted to a 960×540 pinhole image. The official loop does not recover after a tracking loss, so our runner restarts it on consecutive segments; each segment is its own world. For each segment a Sim(3) to the tag path is fitted on *alignment* tag frames, and *validation* tag frames (1 s of every 4 s) never enter the fit. A segment is kept only if its scale is in $[0.05, 3]$, after a tail cut on growing residuals, and if it stays within 0.15 m of the tag/IMU path. The fused path uses SLAM where kept and the tag/IMU path elsewhere, smoothed by a 5-frame moving average. Four numbers decide its value: coverage, survival at grasps (tag-hidden frames within ±1.5 s of a jaw event that have a SLAM pose), error on validation frames, and the jump where two kept segments meet (Sec. VIII-F1). The camera–IMU and camera-to-TCP calibrations it needs are in Sec. V-F.

Jaw state. An ImageNet-pretrained ResNet-18 per hand classifies a fixed region of the wrist fisheye that shows the bare jaws (bottom centre; the left ROI is lower because that camera sees the jaws only in the bottom 30% of the image). The crop is resized to 224×224. Training uses spoken open/closed capture cycles with brightness, contrast, gamma, shift and blur augmentation, AdamW, and validation on held-out cycles. The output p_{open} is thresholded with hysteresis: right 0.3/0.1, left 0.6/0.4. The left images are washed out, and closed frames

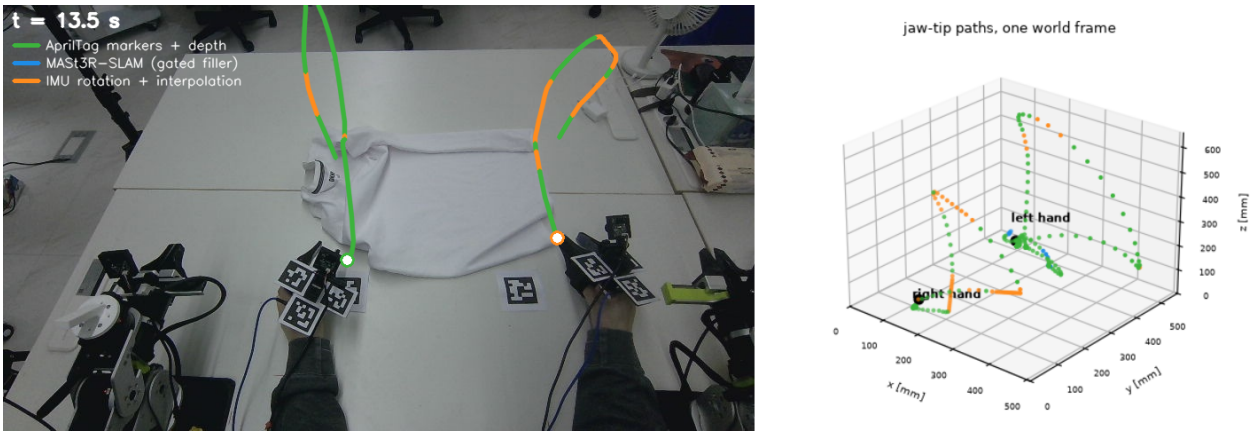


Fig. 5. Reconstructed jaw-tip trajectories D4 of demo 2 (take 171057), one frame of the animation in `media/`. Left: the last 3 s of both paths projected into the Orbbec image, coloured by the source of each frame: AprilTag markers + depth (green), MASt3R-SLAM filler (blue), IMU rotation + interpolation while the markers are hidden (orange). Right: both paths up to this time in the robot base frame (offset to the workspace corner). Orange appears exactly when a hand grasps or lifts the cloth.

score 0.2–0.35; with the right-hand thresholds the left jaw appeared "reversed".

F. Robotization: D4 → D5

Input: D4. *Output:* D5, a robot-feasible jaw-tip trajectory, with the human path kept next to it and a per-frame quality label.

The human path is changed only by continuous warps, never by clipping. Per demo window and hand, let y be the lateral coordinate and d the minimum hand separation (`--sep`, 0.06 m). For the right hand, with y_o and y_i the 1st and 99th percentiles of y,

$$s = \begin{cases} 1, & y_i \le -d/2, \\ \mathrm{clip}\big(\frac{-d/2-y_o}{y_i-y_o},\, 0.4,\, 1\big), & \text{otherwise}, \end{cases} \tag{10}$$

$$y' = \mathrm{softceil}\big(y_o + s\,(y - y_o),\ -d/2,\ w\big), \tag{11}$$

and the left hand is mirrored. The height gets a soft floor,

$$z' = \mathrm{softfloor}(z,\, z_\mathrm{t} + c,\, w), \tag{12}$$

$$\mathrm{softfloor}(z, f, w) = f + w \log\big(1 + e^{(z-f)/w}\big), \tag{13}$$

with $\mathrm{softceil}(y, c, w) = -\mathrm{softfloor}(-y, -c, w)$, $w = 10$ mm and clearance $c = 0$. The softplus floor also lifts points that lie exactly at the floor by $w \ln 2 \approx 7$ mm. The y compression addresses hands that cross the body midline and come within 3–11 cm of each other (seen in take 113243); the floor addresses tag TCPs below the table. An optional minimum reach from each arm base is off.

The displacement $\delta = \|p'_B - p_B\|$ of each frame is stored with a quality label: 0 for $\delta < 20$ mm, 1 for <50 mm, 2 otherwise. Training will use the robot-feasible path with these labels as weights; large shifts in contact phases are the harmful ones.

G. Limit-aware IK: D5 → D6

Input: D5, the jaw state of D4 and the robot model with the overrides of Sec. V-D. *Output:* D6, a joint trajectory for the 2×6 joints plus jaw commands at 30 Hz.

Target. For each command sample the target is

$$T^\star = \Big(R_{B,\mathrm{tcp}}(t)\, R_{UE}\, R_z(\pi)^f\, R_z(\varphi)\ ,\ p'_B(t) \Big), \tag{14}$$

where R_{UE} is the UMI-to-robot-gripper orientation (Sec. V-F), $f \in \{0, 1\}$ the jaw flip and φ the relaxed wrist roll (below). The human time is resampled to the command rate (30 Hz) at the replay speed, and positions are smoothed by a 15-sample moving average on that grid.

IK. Pink [25] solves a differential IK QP per iteration with a frame task on the TCP (position cost 1, orientation cost 0.1, so position comes first), a weak posture task toward the seed (cost 10^{-3}), and the joint limits of the bridge, shifted by the joint offsets and tightened by 0.5°.

- *Step bound.* The first frame is solved freely from READY; for every later frame the accumulated change of each joint is box-projected to $|q_j - q_j^{\mathrm{prev}}| \le 5°$. Without this bound, j6 reaching its $\pm 130°$ limit made the IK jump to another solution (2500°/s).
- *Wrist-roll relaxation.* j6 turns about the tool axis, on which the TCP lies, so a roll φ about that axis moves j6 by about φ without moving the TCP. With threshold $\theta = 115°$ and q_6 the solved j6,

$$\varphi \leftarrow \begin{cases} \varphi - \mathrm{sgn}(q_6)\,(|q_6| - \theta), & |q_6| > \theta, \\ \varphi - \mathrm{sgn}(\varphi)\,\min\big(|\varphi|, 1°, \theta - |q_6|\big), & \text{otherwise}. \end{cases} \tag{15}$$

  Position is never traded.
- *Jaw symmetry.* The parallel jaw is symmetric under 180° about the tool axis. Both $f = 0$ and $f = 1$ are planned, and the one with fewer frames above 10 mm IK error (then the smaller maximum error) is used.
- *Finger-corner clearance.* With x_E, y_E the tool axes, the lowest corner of the fingers is estimated as

$$z_\mathrm{low} = z_\mathrm{tcp} - h\,|x_{E,z}| - (a + b\,g)\,|y_{E,z}|, \tag{16}$$

  with $h = 20$ mm (half finger length along the tool x), $a = 8$ mm, $b = 28$ mm and g the jaw opening from the

classifier. If $z_{\text{low}} < z_{\text{t}} + c_{\text{tip}}$, the target is raised by the difference. The default is $c_{\text{tip}} = +5$ mm (from the first cloth grasp, Sec. VIII-C); the full two-arm replays use $c_{\text{tip}} = -10$ mm (Sec. VIII-E).
- *Floor.* Targets below $z_{\text{t}} + 10$ mm are raised to it and counted.

The joint path is smoothed by a 9-sample moving average. Approach and retreat legs connect READY to the path in Cartesian space (READY → above the start → start, end → above the end → READY; the short descent onto the start is a joint-space line); a joint-space line between far poses had taken links under the table.

Jaw commands. At 600°/s jaw speed the robot jaw reaches half travel 331–337 ms after its command (no load; at the earlier 90°/s it took 1969 ms). Jaw commands are therefore placed before the human event: open by 320 ms and close by 480 ms. These leads were set from a gripper-only replay so that the robot jaw is at 50% when the human jaw is at mid-stroke (the classifier's open event fires near mid-stroke, its close event about 150 ms after it), and verified on a second run (robot 50% at the open event $+4$ ms, at the close event -171 ms for a target of -150 ms). At replay speed s, a lead L in robot time is a lead sL in human time.

H. Safety checks: D6 → approved plan

Input: D6. *Output:* the same plan, approved for execution, or a rejection.

The plan must pass all of these before `--go`: maximum IK position error and rotation deviation, joint limits, joint speed, TCP height, height of links 3–6, gripper and fingers above the table, and a minimum distance of 120 mm between any link origin of the left arm and any of the right arm (every 5th step and along the approach and retreat legs). A regression test replans the right-arm grasp window of the first cloth grasp and fails if IK error, joint speed, link clearance or arm gap get worse.

I. Execution: approved plan → D7

Input: the approved plan. *Output:* D7, the robot motion (commanded and encoder joints) on the host clock.

Commands stream at 30 Hz to the 100 Hz MIT-mode bridge (Sec. IV-C); at speed 0.35 a 33 s demo takes 94 s. The log stores, for each command, its human source time, so every robot sample can be mapped back to the demonstration.

VII. Experimental Setup

A. Data and runs

- *Single-arm parity* (10-20): stroke takes with six translation directions, rotation strokes and free combined motion, replayed on one arm.
- *Take 113243*, demo 1: the first cloth grasp (right arm, grasp window 19.5–24 s), the robotization example of crossing hands, and the SLAM tests.
- *Take 171057*, demo 2 (33 s): the full two-arm replay (10-21) and the SLAM tests. Conversion without hand edits: tag + depth pose, MASt3R fill where it agrees, robotization with $d = 0.06$ m and $c = 0$. Runs: 190154 (stopped early), 193127 (planner default $c_{\text{tip}} = +5$ mm), then 194526, 195154, 195745 and 200335 with one unchanged plan ($c_{\text{tip}} = -10$ mm).

B. Metrics and parity gates: D7 vs. D4/D5

The replay is compared at three levels: A is the human jaw-tip TCP from tags (D4), B the robot command (FK of the commanded joints of D6), C the robot (FK of the encoder readings of D7). When the replayed path was robotized (the two-arm replay), the reference is the robotized path A′ (D5) that was replayed instead of A. For a free-motion window:

- A is taken relative to the median pose of the hold before the motion, B and C relative to their first sample;
- C is shifted in time by the lag $L^\star$ that minimises the mean of $\|C(t+L) - B(t)\|^2$ (searched over 0–1.5 s in 20 ms steps), so that lag is not counted as error;
- position error $e_p(t) = \|C(t+L^\star) - A(t)\|$, reported as RMSE and p95; rotation error is the geodesic angle between the relative rotations of C and A, reported as median and p95;
- amplitude ratios (RMS distance from the start, C over A), C−B and B−A RMSE, and a count of jumps.

Gates. RMSE <20 mm, p95 <30 mm, rotation median <5° and p95 <10°, ratios 0.9–1.1, no jumps. Single-axis strokes are compared on normalised progress along A's chord instead, with gates on the main-axis ratio (0.9–1.1), cross-axis leakage (<10%), endpoint (<20 mm), RMSE and p95. Jaw timing is compared on the normalised phase of the window (UMI event, command, encoder crossing). For the full two-arm replay the whole demo is one window.

C. Video alignment

The same Orbbec films the replay, and the stored mapping from command time to human time produces frame-aligned side-by-side videos (Fig. 1), in human time or, with `--robot-time`, in robot time.

VIII. Results

A. Calibration accuracy

Table II gives the calibration results. The camera-to-robot solve reaches 0.98 px and 1.3 mm lateral error on held-out poses (18 of 32 poses used). The tag rig places the jaw tip with 1–4 mm static error; leaving one board corner out gives 1.8–4.8 mm, and the independent pivot take leaves a residual RMS of 6.7 mm (right) and 5.3 mm (left). The absolute jaw position of the corrected robot model agrees with tape measurements within 6–9 mm at READY, and the left j2 fit within 5–10 mm at its three poses (residuals ≤8 mm, RMS 4.8 mm). The UMI-to-robot-gripper orientation has a spread of 2.3° (right) and 3.8–8.6° (left).

B. Single-arm parity (10-20)

- Six-direction translation strokes: 16/18 pass; median RMSE X 11.6, Y 7.3, Z 5.5 mm (max 14.5 mm); robot endpoint repeatability $\leq$8 mm. The two failures were a marginal cross-axis leak (11.1%) and one bad human stroke.
- Gain history: earlier settings, starting from K_p 64 and K_i 28, had passed 0/6, 3/6 and 5/6 strokes. Raising K_i from 28 to 56 on all joints, then to 84 on j2, reduced the X-stroke RMSE from 25/37 mm to 17/16 mm and then to 14/12 mm (Sec. IV-C).
- Rotation strokes: ratio 0.98–1.04, geodesic median 0.3–1.6°, repeat $\leq$1.4°.
- Free combined motion at 0.9×: 2/2 pass (RMSE 13.7/14.6 mm, p95 26/29 mm, rotation median 1.7/1.4°, ratios 0.99–1.03, robot lag about 0.2 s).
- In these tests B−A was at most 3.4 mm; the error was robot tracking (C−B).
- The jaw command follows the human jaw event within 0.5% of progress.

C. First cloth grasp (10-21)

After the TCP fix (Sec. V-D), the right arm replayed the grasp window of demo 1 of take 113243 and picked up and lifted the T-shirt. The first attempt scraped the table at a 46° tool tilt, which led to the finger-corner clearance with $c_{\mathrm{tip}} = +5$ mm (Sec. VI-G).

D. Two-arm replay of a full demo (10-21)

Input. In the demo window of take 171057, demo 2, the right hand had at least one tag in 79% of the frames and the left hand in 95%; 21% of the right-hand frames had neither a tag nor a kept SLAM pose and are interpolated. Robotization did not compress y in this demo (each hand stayed on its side); only the table floor acted. The per-frame displacement had a median of 0.1 mm (right) and 1.1 mm (left), a p95 of 9.3 and 20.6 mm, and a maximum of 10.5 and 21.5 mm; all right-hand frames and 92% of the left-hand frames are in quality class 0, the rest in class 1 (Fig. 6).

Plan. IK error at most 6.3 mm/11.5° (right) and 10.3 mm/21.0° (left), joint speed $\leq$88°/s, lowest TCP target 13 mm (commanded TCP 4 mm after the joint smoothing), lowest link 1–5 mm above the table, arm-to-arm link distance $\geq$269 mm.

Error against the human (Table III): 17 mm RMSE right and 30 mm left against the replayed, robotized path A′. Against the raw human path (D4) the same runs give 22 mm (right) and 32–34 mm (left), because the robotization displacement (Fig. 6) then also counts as error. The four runs agree within about 1 mm, so the robot is deterministic and the remaining error is systematic.

- *Where the left error comes from:* the left command already deviates from A′ (B−A′ 22.6 mm, right 8.6 mm); the left IK gives up to 21° of orientation. Against the raw human path, the larger left robotization adds to this.
- *Tracking:* the robot follows its own command with 12–16 mm RMSE over the path (C−B), and closely at the grasp moments (Sec. VIII-E, Fig. 7).

E. Jaw height at the grasp

In the first full replays (run 193127, $c_{\mathrm{tip}} = +5$ mm) the jaws closed 19–61 mm above the table and missed the cloth. The cause was the finger-corner clearance: with the tool tilted, the open fingers' lowest corner is 20–36 mm below the TCP, so the planner lifted the TCP.

- *Allowing the corner to touch* ($c_{\mathrm{tip}} = -10$ mm), as the human's fingers do, moves the near-table closes to 14–16 mm (right, 4 closes) and 5–19 mm (left, 5 closes), with an xy error of 2–8 mm against the command at that moment (run 194526, Fig. 8). Every link stays at least 1 mm above the table.
- *Tilting the tool toward vertical instead* (only where the corner would touch) was tried: the IK could not follow even 15° (errors up to 100 mm).
- *Closing at the lowest point* nearby, a heuristic, was rejected.

Outcome. In runs 195154 and 195745 the shirt lay 15–20 cm farther away than in the demo, and the jaws closed on the empty table. In run 200335 the jaws caught the shirt edge and dragged it (Fig. 1); the shirt was moved, but not folded as by the human. The earlier full run 190154 stopped when the right arm lost power; its right-arm encoder log ends 52 s into the 94 s replay.

F. Negative results

1) SLAM as the trajectory source: The tags are hidden exactly when the hand grasps, palm down, so wrist-camera SLAM was tested as the main path.

- *MASt3R-SLAM* (restart runner of Sec. VI-E, Table IV): where it tracks it is accurate (2–14 mm on validation frames), but it keeps only 19–48% of the frames and 0–38% of the tag-hidden frames around grasps. White cloth fills the view at close range and the hand moves fast; each restart starts a new world that needs tag anchors. On the earlier UMIalign takes it also drifted within segments: a single Sim(3) per segment left a median residual of 69 mm (9 mm with 2 s windows), and the tilt drifted 5–12° over 10 s.
- *ORB-SLAM3 mono-inertial* [18] (take 113243, both hands): initialised and gave accurate rotation, but the translation scale diverged 30–1000× (coverage 0–11%, grasp survival 0%). The hand's acceleration (median 0.07–0.16 m/s^2) is below the accelerometer bias (0.25–0.43 m/s^2), so the IMU cannot observe scale; white cloth at close range adds weak visual constraints.
- *Stereo VIO* (10-15/16, an earlier stereo camera): AirSLAM stereo-inertial diverged (15 m range, 21% lost frames); stereo VO alone ran with 3.6% lost frames; Basalt and OpenVINS were also set up. The stereo camera was dropped on 10-16.

SLAM is therefore kept only as a gated filler (Sec. VI-E).

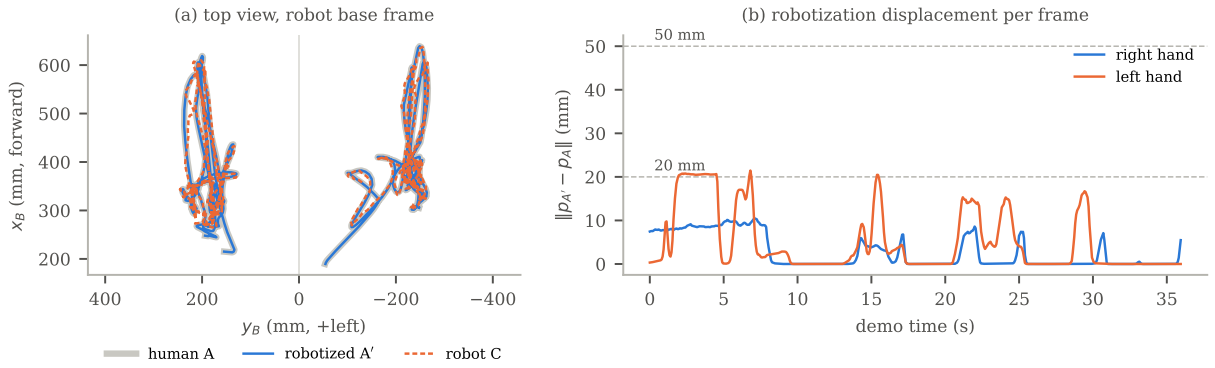


Fig. 6. Demo 2 of take 171057 and run 200335. (a) Top view of the human TCP (A, wide gray), the robotized target (A′) and the robot from encoders (C) for both hands in the robot base frame; A and A′ coincide in xy because only the height was warped. (b) Robotization displacement per frame; the dashed lines are the 20 and 50 mm quality thresholds. Computed from `umi_traj_*_fused_robot.npz` and the replay log.

TABLE III
TWO-ARM REPLAY OF DEMO 2 AGAINST THE REPLAYED, ROBOTIZED HUMAN PATH A′ (D5), PER RUN (WHOLE DEMO AS ONE WINDOW)

run	arm	human range	pos. RMSE	pos. p95	rot. median	rot. p95	endpoint	C−B	B−A′
194526	right	464 mm/ 40°	17.2 mm	36.8 mm	8.1°	12.6°	20 mm/ 9.7°	12.1 mm	8.6 mm
194526	left	572 mm/ 56°	29.8 mm	51.0 mm	12.6°	17.1°	19 mm/ 16.2°	14.5 mm	22.6 mm
195154	right		17.3 mm	36.8 mm	8.1°	12.7°	20 mm/ 9.8°	12.2 mm	8.6 mm
195154	left		30.3 mm	49.5 mm	12.7°	17.0°	19 mm/ 16.0°	15.2 mm	22.6 mm
195745	right		17.2 mm	35.3 mm	8.1°	12.7°	20 mm/ 9.7°	12.1 mm	8.6 mm
195745	left		29.0 mm	49.8 mm	12.3°	17.2°	18 mm/ 16.0°	15.2 mm	22.6 mm
200335	right		17.4 mm	35.8 mm	8.1°	12.6°	20 mm/ 9.7°	12.3 mm	8.6 mm
200335	left		30.3 mm	51.3 mm	12.7°	17.2°	21 mm/ 16.0°	15.5 mm	22.6 mm

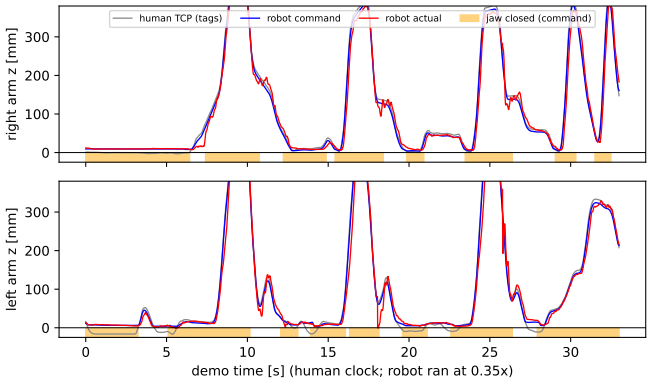


Fig. 7. Height of the human TCP (tags), the robot command and the robot (encoders) over the demo (run 200335). Orange: jaw closed. The robot follows the robotized path closely; near the table the command sits slightly above the human because of the soft table floor.

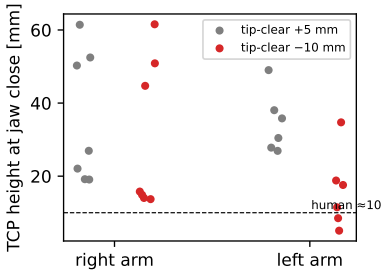


Fig. 8. TCP height at each jaw close, with the default finger-corner clearance (+5 mm, run 193127) and with the corner allowed to touch (−10 mm, run 194526). The dashed line is the human (≈10 mm).

2) Earlier labels (10-13 to 10-16): Before the tag rig, a smoother fused MASt3R, Orbbec body tracking of the UMI and the IMU. It had three errors, all invisible to its own quality checks (Orbbec residual, jumps, coverage):

- a stale Kalibr camera–IMU rotation (17–24° off);
- a free camera-to-TCP lever that converged 20–35 cm wrong (it had to be fixed from the jaw tips seen by the Orbbec);

TABLE IV
MAST3R-SLAM ON THE WRIST VIDEO (RESTART RUNNER)

take / hand	kept	grasp survival	hidden err.	jump
113243 right	19%	0%	2.2 mm	39.6 mm
113243 left	28%	38%	13.9 mm	72.9 mm
171057 right	20%	12%	–	–
171057 left	48%	32%	–	–

- dark sleeves taking over the body anchor (removed by a gate against the top AprilTag).

After fixing them, the TCP was still only 38–59 mm (median

per session) from the jaw tape seen by the Orbbec. A 65-episode "flatten and align" set built from these labels was used to fine-tune a policy; it is not evaluated here. The 4–6 cm label error motivated the 1–4 mm tag rig.

IX. Discussion and Lessons

1) *Absolute tags plus depth beat SLAM for contact-rich hand motion.* The failure regime of SLAM (close range, textureless cloth, fast motion) is exactly the grasp.
2) *Measure in the world, not in FK.* The 89 mm TCP error and the 70 mm base error were invisible to FK-based checks; tape, camera and real cloth found them.
3) *Fix the conversion, do not add heuristics.* Lifting the jaws for finger-corner safety cost the grasp; allowing contact restored it.
4) *One clock for everything.* It made camera, IMU and robot alignment, and the side-by-side videos, possible without sync hardware.
5) *Model the controller.* The tracking error followed $\dot{\tau}_g / K_i$; changing K_p could not help.
6) *Grade the data instead of gating it.* Strict gates leave no usable egocentric data. Three data grades (replay / training / pretraining) and per-frame quality weights keep it usable.

X. Limitations and Threats to Validity

- *No success rate.* Replays were judged by measurement and video; no trials were counted. One demo of one take was replayed in full on both arms, four times with the same plan; the 1 mm repeatability shows determinism, not generality across demos, demonstrators or garments.
- *What the parity metrics cannot see.* A, B and C are compared relative to the start of each window and C is shifted by its fitted lag, so constant offsets and delays do not count as error. C is the FK of the encoders and carries the same kinematic model as B. This is why the 89 mm TCP error passed every parity test. The absolute jaw position was checked only by tape at a few poses (Sec. VIII-A) and by the Orbbec video.
- *Tag-hidden phases.* Position there is interpolated (21% of the right-hand frames of demo 2) or taken from gated SLAM; its error is not measured at the grasps that matter most.
- *Jaw labels.* The classifier thresholds were tuned per hand by inspection; we have no held-out accuracy number on demo takes. The jaw leads were measured without load.
- *Table model.* The table is a plane at $z_{\mathrm{t}} = 3$ mm in the right-arm base frame; the soft floor and clearances depend on it.
- *Video timing.* The Orbbec recording of run 200335 lost its timestamp file (the recorder was suspended), so its side-by-side video uses frame write times (about 10–30 ms error).

XI. Conclusion and Next Steps

Replay of converted hand-worn-gripper demonstrations works as an end-to-end test: with one world frame and one clock, a two-arm robot reproduces a full demonstration deterministically and brings its jaws to the human's grasp height, but it does not yet fold like the human. The open problems and next steps are:

- *Position error* of 17–30 mm RMSE over the demo against the replayed path (22–34 mm against the raw human path); the error at the grasp points themselves is not yet measured. Next, a per-grasp xy error table from the videos: a constant bias points to calibration, a varying one to local visual correction at the grasp.
- *Open-loop replay needs the same scene.* Next, a cloth-placement reference before each replay.
- *Tag-hidden phases.* Next, an offline smoother (tags, IMU, depth and gated MASt3R) with confidence-weighted labels.
- *Hardware.* The left wrist connector is intermittent, and the right arm has dropped power and USB.
- *Training* on relative Cartesian labels of the robotized path, weighted by quality, has not started.

Appendix A
Running a session

Interpreters. `.venv` records and processes (Orbbec needs `sudo`); `xvla-mac` plans and replays; the robot plugin's Python runs the bridge. Tool names are given without paths; options are shortened.

Bring-up (every session). Check the USB location IDs and gripper zeros; `bridge_start --sides left,right --go`; `ready_pose --side right|left --go`.

Calibration (one-time, Sec. V).

- Camera↔robot: `robot_cam_poses --side S --go` (32 holds) → `robot_cam_solve.py`.
- Start markers: result in `configs/calibration/start_anchors_v001.json`.
- Tag rig: `tag_rig_solve.py`, `tag_rig_pivot.py`.

Data chain, in order (Sec. VI).

- →D0 record: `sudo tag_rig_record --proto task --wrist` (start at markers 400/401).
- D0→D3 tags to jaw-tip trajectory: `umi_tcp_traj.py --hand H --depth --imu-fill` for each hand (tag fusion: `tag_rig_fuse.py`); the IMU fill of D3→D4 runs in the same call.
- D3→D4 optional SLAM: `wrist_m3_prep.py` → `run_segmented.py` (GPU) → `umi_slam_eval.py --save`.
- D3→D4 jaw state: `umi_jaw infer` on `wrist/` and `wrist_left/`.
- D4→D5 robotize: `umi_robotize_bi.py --src _fused --clear 0 --sep 0.06 --min-reach 0`.
- D5→D6 + safety checks: `umi_replay_bi --rep N --traj _fused_robot --speed 0.35 --q-smooth 9 --max-ik-mm 15`

`--tip-clear -0.01 --tip-clear-left-extra 0 --link-margin 0.015 --max-rot-dev 25 --offline`.

- Plan→D7 execute: the same with `--go`, then `POST /rest`.
- Compare D7: `replay_parity.py`; `replay_side_by_side.py [--robot-time]`.

Safety. The robot moves only with a person present. Plan offline before `--go`. Park the arms between tests; holding READY for hours overheated a motor. Stop the recorder with Ctrl+C, not Ctrl+Z, so that its timestamps are saved.

APPENDIX B
REPOSITORY LAYOUT

- `technical_report/`: this report (LaTeX, figures, PDF).
- `hardware/umihand/`: HandUMI build copy with our modifications, tag sheets and IMU firmware.
- `visual_teleop/`: recorder, jaw classifier, IK planner and replay, robot tools.
- `scripts/`: offline processing for the current pipeline (tag rig, UMI pose, SLAM evaluation, robotization, parity, side-by-side video).
- `slam/mast3r/`: MASt3R restart runner.
- `robot_bridge/`: reBot MIT bridge and gripper/joint zero tools.
- `configs/`: calibration, robot settings and overrides, jaw weights.
- `media/`: GIFs, videos and photos.
- `legacy/`: earlier and dropped approaches (bare-hand collector, UMIalign smoother and exporter, red-tape jaw tools, Orbbec body tracking, Kalibr export, ORB-SLAM3 and stereo-VIO experiments).

Raw takes, the robot plugin and the URDF are not included. ORB-SLAM3 (GPLv3) is represented only by our patch scripts.

REFERENCES

[1] C. Chi, Z. Xu, C. Pan, E. Cousineau, B. Burchfiel, S. Feng, R. Tedrake, and S. Song, "Universal Manipulation Interface: In-the-wild robot teaching without in-the-wild robots," in *Proc. Robotics: Science and Systems (RSS)*, 2024. arXiv:2402.10329.

[2] BrikHMP18 and murobotics-ai, "HandUMI hardware (handumi-hw)," GitHub repository, Apache-2.0, 2026. https://github.com/murobotics-ai/handumi-hw (build copy from commit `e58de33`).

[3] B. Meza, L. Perez, A. Mendoza, A. Ubillus, M. Ramos, R. Bastidas, M. Olivera, and B. Bastidas, "HandUMI: A modular hand-worn parallel-jaw interface for robot-free bimanual data capture," 2026. https://murobotics.ai/research/handumi

[4] M. Xu, H. Zhang, Y. Hou, Z. Xu, L. Fan, M. Veloso, and S. Song, "DexUMI: Using human hand as the universal manipulation interface for dexterous manipulation," in *Proc. Conf. Robot Learning (CoRL)*, PMLR vol. 305, pp. 437–459, 2025. arXiv:2505.21864.

[5] S. Kareer, D. Patel, R. Punamiya, P. Mathur, S. Cheng, C. Wang, J. Hoffman, and D. Xu, "EgoMimic: Scaling imitation learning via egocentric video," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, pp. 13226–13233, 2025. arXiv:2410.24221.

[6] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," in *Proc. Robotics: Science and Systems (RSS)*, 2023. arXiv:2304.13705.

[7] R. Cadene *et al.*, "LeRobot: State-of-the-art machine learning for real-world robotics in PyTorch," 2024. https://github.com/huggingface/lerobot

[8] Y. Avigal, L. Berscheid, T. Asfour, T. Kröger, and K. Goldberg, "SpeedFolding: Learning efficient bimanual folding of garments," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2022. arXiv:2208.10552.

[9] H. Ha and S. Song, "FlingBot: The unreasonable effectiveness of dynamic manipulation for cloth unfolding," in *Proc. Conf. Robot Learning (CoRL)*, 2021. arXiv:2105.03655.

[10] E. Olson, "AprilTag: A robust and flexible visual fiducial system," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, pp. 3400–3407, 2011.

[11] J. Wang and E. Olson, "AprilTag 2: Efficient and robust fiducial detection," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, pp. 4193–4198, 2016.

[12] G. Bradski, "The OpenCV library," *Dr. Dobb's Journal of Software Tools*, 2000.

[13] T. Collins and A. Bartoli, "Infinitesimal plane-based pose estimation," *Int. J. Computer Vision*, vol. 109, no. 3, pp. 252–286, 2014.

[14] G. Terzakis and M. Lourakis, "A consistently fast and globally optimal solution to the perspective-n-point problem," in *Proc. European Conf. Computer Vision (ECCV)*, pp. 478–494, 2020.

[15] R. Murai, E. Dexheimer, and A. J. Davison, "MASt3R-SLAM: Real-time dense SLAM with 3D reconstruction priors," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2025. arXiv:2412.12392.

[16] V. Leroy, Y. Cabon, and J. Revaud, "Grounding image matching in 3D with MASt3R," in *Proc. European Conf. Computer Vision (ECCV)*, 2024. arXiv:2406.09756.

[17] S. Wang, V. Leroy, Y. Cabon, B. Chidlovskii, and J. Revaud, "DUSt3R: Geometric 3D vision made easy," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024. arXiv:2312.14132.

[18] C. Campos, R. Elvira, J. J. Gómez Rodríguez, J. M. M. Montiel, and J. D. Tardós, "ORB-SLAM3: An accurate open-source library for visual, visual-inertial, and multimap SLAM," *IEEE Trans. Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.

[19] V. Usenko, N. Demmel, D. Schubert, J. Stückler, and D. Cremers, "Visual-inertial mapping with non-linear factor recovery," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 422–429, 2020.

[20] P. Geneva, K. Eckenhoff, W. Lee, Y. Yang, and G. Huang, "OpenVINS: A research platform for visual-inertial estimation," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, pp. 4666–4672, 2020.

[21] K. Xu, Y. Hao, S. Yuan, C. Wang, and L. Xie, "AirSLAM: An efficient and illumination-robust point-line visual SLAM system," *IEEE Trans. Robotics*, 2025 (accepted). arXiv:2408.03520.

[22] P. Furgale, J. Rehder, and R. Siegwart, "Unified temporal and spatial calibration for multi-sensor systems," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, pp. 1280–1286, 2013.

[23] M. Shah, "Solving the robot-world/hand-eye calibration problem using the Kronecker product," *J. Mechanisms and Robotics*, vol. 5, no. 3, 2013.

[24] J. Carpentier, G. Saurel, G. Buondonno, J. Mirabel, F. Lamiraux, O. Stasse, and N. Mansard, "The Pinocchio C++ library: A fast and flexible implementation of rigid body dynamics algorithms and their analytical derivatives," in *Proc. IEEE/SICE Int. Symp. System Integration (SII)*, 2019.

[25] S. Caron *et al.*, "Pink: Python inverse kinematics based on Pinocchio," software, Apache-2.0. https://github.com/pink-kinematics/pink

[26] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016.

[27] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 248–255, 2009.